Feature-Based and Amino-Acid-Based ClonalTree Report

Ground-Truth Validation and Alternative Representations for B-Cell Lineage Reconstruction

Ongoing Research

Pouya Mirzaeizadeh
Shadi Sefidgar

August 21, 2026

Contents

Project Status

Important Notice

This repository documents ongoing research on alternative approaches for B-cell lineage reconstruction using the ClonalTree framework.

The project is currently under active development and has not yet been completed. Some implementation details, algorithms, and experimental analyses are intentionally omitted because the work is ongoing and is being prepared for future publication.

The purpose of this report is to document the motivation, methodology, validation experiments, current results, and future directions of the project.

The current work investigates alternative representations for sequence similarity while preserving the overall ClonalTree reconstruction framework. The main approaches considered are:

1. traditional nucleotide-based reconstruction;
2. amino-acid-based reconstruction;
3. numerical feature-based reconstruction;
4. biologically informed reconstruction based on chemical and physicochemical properties of amino acids.

A ground-truth validation experiment was also introduced to establish a quantitative baseline for the reconstruction pipeline before comparing the alternative sequence representations.

1 Introduction

B-cell lineage reconstruction is an important problem in computational immunology. During affinity maturation, B-cells undergo somatic hypermutation, generating populations of related antibody sequences. Reconstructing the evolutionary relationships among these sequences can provide insight into clonal evolution, immune responses, and antibody development.

A lineage tree represents the inferred relationships between observed sequences and their potential ancestors. In a correctly reconstructed tree, sequences that are evolutionarily or biologically related should generally be placed within the same lineage or sublineage.

Many existing lineage reconstruction methods rely primarily on nucleotide-level sequence similarity. ClonalTree is one such framework that combines sequence distance, abundance information, and Minimum Spanning Tree (MST) construction to infer lineage relationships efficiently.

The use of nucleotide-level sequence similarity provides several advantages. Nucleotide sequences contain detailed information about the underlying mutational history and can be compared using simple distance measures such as Hamming distance. However, nucleotide-level similarity does not necessarily correspond directly to biological or functional similarity at the protein level.

For example, a nucleotide substitution may be synonymous and therefore leave the encoded amino acid unchanged. Conversely, two different amino-acid substitutions may contribute equally to a simple amino-acid Hamming distance even though their chemical and structural consequences may be very different.

These observations motivate the investigation of alternative representations for lineage reconstruction.

The objective of this research is to investigate whether alternative sequence representations can provide biologically meaningful lineage reconstruction while maintaining the computational efficiency of the ClonalTree framework.

In particular, three main directions are investigated:

1. traditional nucleotide-based lineage reconstruction;
2. amino-acid-based lineage reconstruction using protein-level sequence similarity;
3. feature-based and biologically informed reconstruction methods that incorporate numerical, chemical, and physicochemical properties.

The study follows two complementary evaluation strategies.

First, simulated lineage trees with known ground-truth topologies are used to quantitatively evaluate the reconstruction pipeline using Precision, Recall, and F-measure.

Second, reconstructed trees are visually compared to examine differences in branching structure and topology between nucleotide-based, amino-acid-based, and feature-based representations.

The long-term goal is to determine how the choice of sequence representation affects reconstruction accuracy, lineage topology, biological interpretability, and computational efficiency.

2 Nucleotide-Based ClonalTree Reconstruction

2.1 Overview

The nucleotide-based ClonalTree implementation serves as the baseline reconstruction method for this study.

The baseline uses aligned nucleotide sequences to calculate pairwise sequence distances. These distances are then supplied to the existing ClonalTree Minimum Spanning Tree procedure.

The purpose of establishing this baseline is twofold. First, it provides a reference implementation against which alternative representations can be compared. Second, it allows the reconstruction pipeline itself to be validated using simulated ground-truth trees.

2.2 Sequence Representation

For the nucleotide-based reconstruction, each sequence is represented by its aligned nucleotide sequence.

For two sequences i and j, the Hamming distance is calculated as

$$d(i,j) = \sum_{k=1}^{L} \mathbb{I}(x_{i,k} \neq x_{j,k}),$$

where L is the aligned sequence length and $\mathbb{I}$ is an indicator function that takes the value 1 when the nucleotides differ and 0 otherwise.

The resulting pairwise distances form an adjacency or distance matrix:

$$D = \begin{bmatrix} 0 & d(1,2) & \cdots & d(1,n) \\ d(2,1) & 0 & \cdots & d(2,n) \\ \vdots & \vdots & \ddots & \vdots \\ d(n,1) & d(n,2) & \cdots & 0 \end{bmatrix}.$$

This matrix is then supplied to the standard ClonalTree MST reconstruction procedure.

2.3 Minimum Spanning Tree Reconstruction

The ClonalTree framework constructs a Minimum Spanning Tree using the calculated sequence distances together with sequence labels, abundance information, and the specified root.

The core reconstruction procedure follows the general form

```
primMST(adjMatrix, root, labels, abundance).
```

The resulting tree is subsequently processed using trimming and revision procedures when required.

The important aspect of this implementation is that the underlying tree construction algorithm is not modified when alternative distance representations are introduced. Instead, the main experimental variable is the representation used to calculate the pairwise distance matrix.

This makes it possible to investigate whether changes in sequence representation alone can affect lineage reconstruction.

3 Ground-Truth Validation

3.1 Motivation

A major difficulty in evaluating lineage reconstruction methods is that the true evolutionary history of experimentally observed B-cell populations is usually unknown.

If the true lineage tree is unavailable, a reconstructed topology can be inspected visually, but it is difficult to determine quantitatively how many of its relationships are correct.

To address this problem, simulated datasets with known lineage topologies were generated using Anestra Eco.

These simulations provide a controlled environment in which the reconstructed trees can be compared directly with their corresponding ground-truth trees.

Four simulated lineage trees were generated for the initial validation experiment.

The primary objective was to determine how accurately the nucleotide-based ClonalTree implementation can recover these known topologies.

The same evaluation framework can then be applied to the amino-acid-based implementation, allowing the two approaches to be compared under identical conditions.

3.2 Generation of Ground-Truth Datasets

Four simulated lineage trees with known topologies were generated using Anestra Eco.

Each simulated tree represents a known lineage history and provides the corresponding sequence dataset used for reconstruction.

The four datasets differ in size and structural complexity, allowing the reconstruction pipeline to be tested under different conditions.

The complete validation workflow was:

1. Generate a simulated lineage tree using Anestra Eco.
2. Obtain the corresponding sequence dataset.
3. Reconstruct the lineage using nucleotide-based ClonalTree.
4. Compare the reconstructed topology with the original ground-truth topology.
5. Remove additional inferred intermediate nodes when necessary.
6. Compare the resulting edge sets.
7. Calculate Precision, Recall, and F-measure.

3.3 Handling Additional Intermediate Nodes

The reconstructed nucleotide trees occasionally contained additional inferred intermediate nodes that were not explicitly present in the simulated ground-truth trees.

A direct comparison between the two topologies would therefore introduce differences caused by the representation of intermediate nodes rather than by incorrect lineage relationships.

To make the comparison fair, additional inferred intermediate nodes were removed before the edge-based evaluation.

This preprocessing step ensures that the evaluation focuses on comparable lineage relationships between the reconstructed and ground-truth topologies.

3.4 Evaluation Metrics

The comparison was performed using edge-level Precision, Recall, and F-measure.

For each reconstructed tree, edges were classified into the following categories:

- **Common edges**: edges present in both the reconstructed and ground-truth trees.
- **False-positive edges**: edges present in the reconstructed tree but absent from the ground-truth tree.
- **False-negative edges**: edges present in the ground-truth tree but absent from the reconstructed tree.

Precision measures the proportion of reconstructed edges that correspond to ground-truth edges:

$$\text{Precision} = \frac{\text{Common Edges}}{\text{Common Edges} + \text{False Positives}}.$$

Recall measures the proportion of ground-truth edges recovered by the reconstruction:

$$\text{Recall} = \frac{\text{Common Edges}}{\text{Common Edges} + \text{False Negatives}}.$$

The F-measure is the harmonic mean of Precision and Recall:

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

The F-measure is particularly useful here because it provides a single value that reflects both the ability to recover correct relationships and the ability to avoid introducing incorrect relationships.

3.5 Nucleotide-Based Validation Results

The nucleotide-based ClonalTree reconstruction achieved high agreement with the four simulated ground-truth trees.

Table 1: Edge-based evaluation of nucleotide-based ClonalTree reconstruction against the simulated ground-truth trees.

Dataset	Tree Edges	Common Edges	FP	FN	Precision	Recall	F-measure
Tree 1	46	45	1	1	0.9783	0.9783	0.9783
Tree 2	82	77	5	5	0.9390	0.9390	0.9390
Tree 3	337	309	28	28	0.9169	0.9169	0.9169
Tree 4	32	31	1	1	0.9688	0.9688	0.9688

The results show consistently high reconstruction accuracy across all four simulated datasets.

The F-measure values range from 0.9169 to 0.9783. The mean F-measure across the four datasets is approximately

$$\frac{0.9783 + 0.9390 + 0.9169 + 0.9688}{4} = 0.95075,$$

or approximately

$$\boxed{F_1 = 0.9508}.$$

This indicates that the nucleotide-based ClonalTree implementation recovers approximately 95% of the evaluated edge relationships on average across the four simulations.

3.5.1 Tree 1

Tree 1 achieved the highest reconstruction accuracy, with an F-measure of 0.9783.

The ground-truth tree contained 46 edges, of which 45 were recovered by the reconstruction. Only one false-positive edge and one false-negative edge were identified.

The false-positive edge was

`seq87--seq88`,

while the missing ground-truth edge was

`seq1--seq88`.

The very small number of discrepancies explains the high Precision and Recall of 0.9783.

This result indicates that the nucleotide-based reconstruction is highly accurate for this simulated topology.

3.5.2 Tree 2

Tree 2 contained 82 ground-truth edges, of which 77 were recovered.

Five false-positive and five false-negative edges were identified, resulting in an F-measure of 0.9390.

The false-positive edges were:

- `seq17--seq18`
- `seq192--seq47`
- `seq20--seq39`
- `seq20--seq40`
- `seq222--seq223`

The false-negative edges were:

- `naive--seq17`
- `naive--seq39`
- `naive--seq40`
- `seq192--seq50`
- `seq223--seq6`

Several of these differences involve relationships near the naive/root sequence. This suggests that some of the reconstruction differences may be associated with local decisions around ancestral or highly connected sequences.

Nevertheless, the reconstruction successfully recovered 77 of the 82 ground-truth edges.

3.5.3 Tree 3

Tree 3 represents the most complex of the four simulated datasets.

It contains 337 ground-truth edges, substantially more than the other datasets. The reconstruction recovered 309 common edges while producing 28 false-positive and 28 false-negative edges.

The resulting F-measure was 0.9169.

This was the lowest F-measure among the four datasets, but it remains above 0.91.

The increased number of discrepancies is likely related to the substantially larger number of edges and the greater structural complexity of the tree. With more lineage relationships, there are more opportunities for local topological differences to occur.

Examples of false-positive edges include:

- `seq15--seq16`
- `seq150--seq38`
- `seq151--seq38`
- `seq176--seq87`
- `seq177--seq87`
- `seq192--seq193`
- `seq197--seq97`
- `seq3--seq902`
- `seq30--seq962`
- `seq371--seq737`

These represent the first ten false-positive examples; the remaining 18 false-positive edges are omitted from the main report for readability.

Examples of false-negative edges include:

- `naive--seq444`
- `seq105--seq418`
- `seq11--seq193`
- `seq11--seq197`
- `seq11--seq388`
- `seq11--seq767`
- `seq11--seq768`
- `seq11--seq791`
- `seq11--seq792`
- `seq117--seq911`

Again, only the first ten examples are shown, with the remaining 18 omitted for readability.

Despite these differences, the reconstruction still recovered 309 of the 337 ground-truth edges.

3.5.4 Tree 4

Tree 4 contained 32 ground-truth edges and 31 common edges.

Only one false-positive and one false-negative edge were identified.

The additional edge was

`seq155--seq156`,

while the missing ground-truth edge was

`naive--seq156`.

The resulting F-measure was 0.9688.

This result is comparable to Tree 1 and indicates that the nucleotide-based method can accurately recover relatively smaller lineage structures.

3.6 Interpretation of the Nucleotide Validation

Overall, the ground-truth experiment demonstrates that the nucleotide-based ClonalTree implementation provides a strong reconstruction baseline.

The method achieved an average F-measure of approximately 0.9508 across the four simulated datasets.

The results are particularly important for the remainder of the project because they demonstrate that differences observed after changing the distance representation can be interpreted relative to a validated baseline.

Tree 3 provides an important example of the relationship between dataset complexity and reconstruction accuracy. Although it produced the lowest F-measure, it was also considerably larger than the other datasets.

Therefore, the reduction from 0.9783 on Tree 1 to 0.9169 on Tree 3 should not necessarily be interpreted as a failure of the method. Instead, it indicates that larger and more structurally complex lineage trees provide a more challenging reconstruction problem.

The results also show that the current edge-based evaluation produces equal numbers of false positives and false negatives for each dataset. As a result, Precision and Recall are identical in the reported experiments.

The F-measure therefore provides a convenient summary of the current evaluation.

3.7 Visual Comparison of Ground-Truth and Reconstructed Trees

Quantitative edge-based evaluation provides an objective measure of reconstruction accuracy, but visual inspection remains useful for understanding how topological differences are distributed throughout each tree.

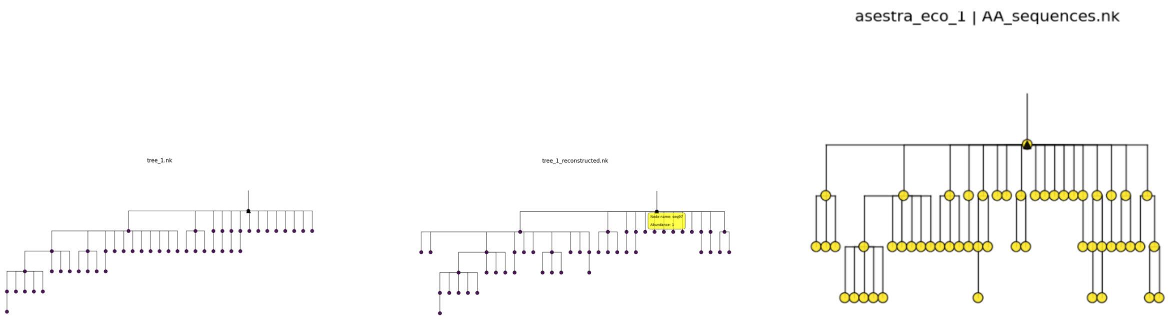
For this purpose, the ground-truth tree generated by Anestra Eco and the corresponding nucleotide- and amino-acid-based ClonalTree reconstructions are presented together for each dataset.

Rather than presenting twelve separate figures, the three representations are combined into four three-panel figures.

Each figure follows the same order:

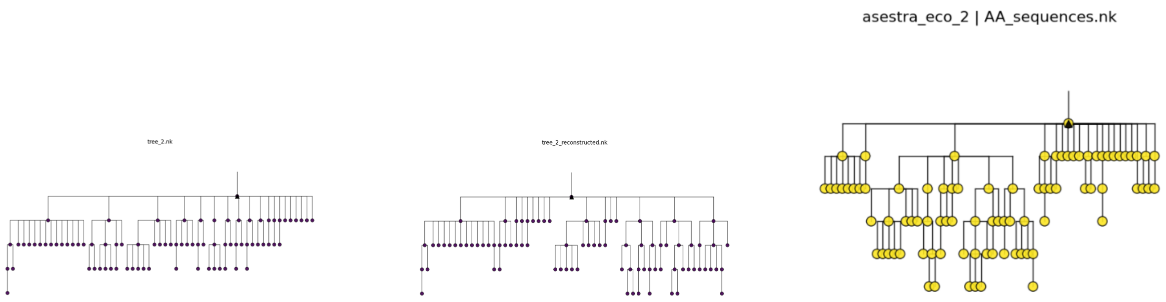
Ground Truth → DNA ClonalTree → AA ClonalTree

This arrangement allows the reader to compare the same dataset across the three representations directly.



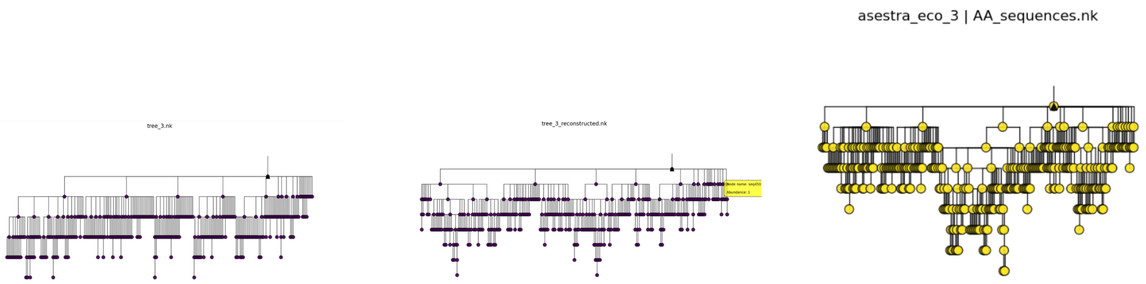
(a) Ground truth (b) DNA (c) Amino acid

Figure 1: Visual comparison of the ground-truth, nucleotide-based, and amino-acid-based lineage trees for Dataset 1.



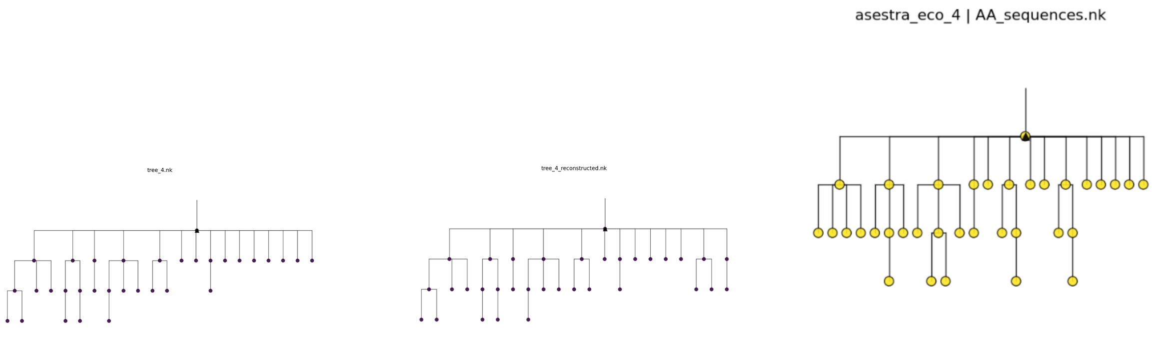
(a) Ground truth (b) DNA (c) Amino acid

Figure 2: Visual comparison of the ground-truth, nucleotide-based, and amino-acid-based lineage trees for Dataset 2.



(a) Ground truth (b) DNA (c) Amino acid

Figure 3: Visual comparison of the ground-truth, nucleotide-based, and amino-acid-based lineage trees for Dataset 3.



(a) Ground truth (b) DNA (c) Amino acid

Figure 4: Visual comparison of the ground-truth, nucleotide-based, and amino-acid-based lineage trees for Dataset 4.

The visual comparison provides an additional perspective on the reconstruction results. In particular, it allows differences in branching patterns, intermediate nodes, and sublineage organization to be examined directly.

The nucleotide-based trees can be quantitatively evaluated using the ground-truth metrics described above.

The amino-acid-based trees are currently included as a visual comparison. Their quantitative evaluation using the same ground-truth datasets and edge comparison procedure is part of the ongoing work.

4 Amino-Acid-Based Reconstruction

4.1 Motivation

Traditional ClonalTree reconstruction relies on nucleotide-level Hamming distance. Although computationally simple and effective, nucleotide similarity does not always capture biologically meaningful relationships between sequences.

A nucleotide sequence contains information about both synonymous and nonsynonymous substitutions. However, synonymous substitutions do not alter the resulting protein sequence.

Therefore, a nucleotide-level distance can treat two sequences as different even when their encoded proteins are identical.

To investigate whether protein-level information can provide a more biologically meaningful representation of clonal evolution, an amino-acid-based version of the ClonalTree reconstruction pipeline was developed.

4.2 Conversion to Amino-Acid Sequences

The LLC benchmark datasets provide aligned nucleotide sequences but do not directly provide amino-acid datasets suitable for the modified ClonalTree pipeline.

Amino-acid sequences were therefore obtained using the IMGT framework.

The amino-acid representation preserves the alignment of the corresponding protein sequences and provides the input for the protein-level distance calculation.

4.3 Amino-Acid Distance

For two aligned amino-acid sequences i and j, the amino-acid Hamming distance is calculated as

$$d(i,j) = \sum_{k=1}^{L} \mathbb{I}(AA_{i,k} \neq AA_{j,k}).$$

The resulting pairwise distances form the amino-acid distance matrix.

The same ClonalTree MST reconstruction procedure is then applied:

```
primMST(AA_adjMatrix, root, labels, abundance).
```

Trimming and revision procedures are subsequently applied when required.

The important point is that the tree construction procedure remains unchanged. Only the representation used to calculate the sequence distances is modified.

This allows nucleotide- and amino-acid-based reconstructions to be compared without changing the underlying MST algorithm.

4.4 Example Output

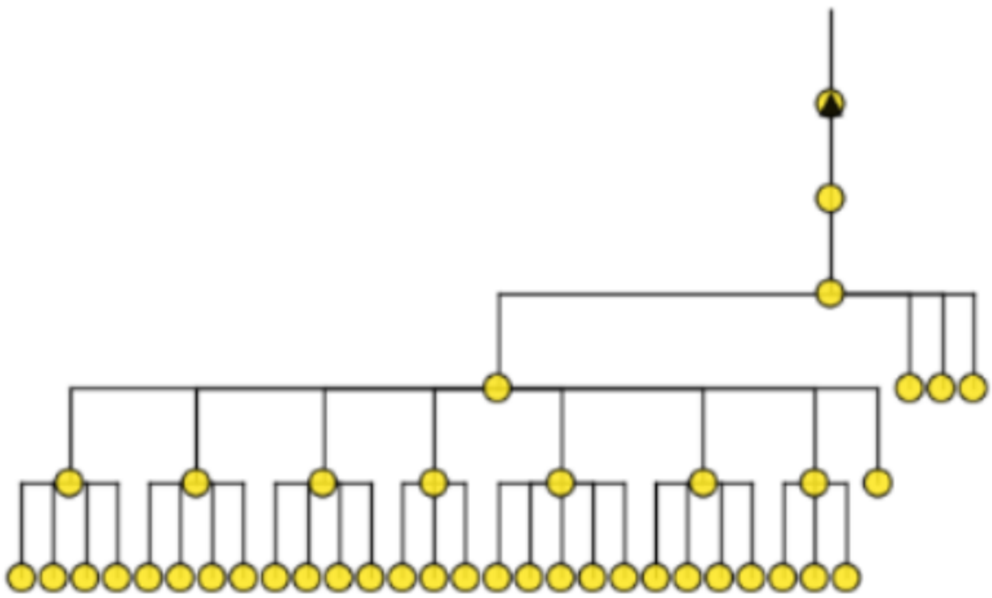


Figure 5: Example amino-acid-based lineage tree generated using ClonalTree and amino-acid Hamming distance.

4.5 Discussion

The amino-acid-based reconstruction produces a structured lineage topology with multiple internal branching points and groups of closely related sequences.

Compared with nucleotide-level representation, amino-acid sequences remove synonymous nucleotide differences and therefore focus the distance calculation on substitutions that alter the encoded protein.

This can provide a different view of the evolutionary relationships among sequences.

Several compact groups of descendant sequences can be observed in the reconstructed trees, connected through common internal ancestors. Such structures are consistent with the presence of related subclones derived from shared ancestral B-cell populations.

However, a visually appealing or highly branched tree does not by itself demonstrate improved reconstruction accuracy.

For this reason, the amino-acid representation must be evaluated against the same simulated ground-truth topologies used for the nucleotide baseline.

4.6 Quantitative Ground-Truth Evaluation

The amino-acid-based implementation will be evaluated using the same edge-based Precision, Recall, and F-measure procedure used for the nucleotide-based reconstruction.

This ensures a direct comparison between the two approaches.

The evaluation will use:

- the same four simulated datasets;
- the same ground-truth topologies;
- the same procedure for handling additional intermediate nodes;
- the same definition of common, false-positive, and false-negative edges;
- the same Precision, Recall, and F-measure calculations.

The current comparison table is therefore:

Table 2: Quantitative comparison of nucleotide- and amino-acid-based reconstruction. Amino-acid evaluation is ongoing.

	Nucleotide			Amino Acid		
Dataset	Precision	Recall	F-measure	Precision	Recall	F-measure
Tree 1	0.9783	0.9783	0.9783	–	–	–
Tree 2	0.9390	0.9390	0.9390	–	–	–
Tree 3	0.9169	0.9169	0.9169	–	–	–
Tree 4	0.9688	0.9688	0.9688	–	–	–

Once the amino-acid evaluation is completed, the difference between the two representations can be quantified directly.

5 Feature-Based Reconstruction

5.1 Initial Implementation

5.1.1 Motivation

The objective of the feature-based implementation was to investigate whether numerical sequence descriptors could be used instead of traditional nucleotide or amino-acid sequence distances for B-cell lineage reconstruction.

Rather than computing distances directly from the sequence characters, each sequence is represented by numerical features extracted from the sequence.

The resulting feature-based distances are then supplied to the standard ClonalTree reconstruction pipeline.

In this way, the overall tree construction procedure remains unchanged while the representation used to calculate pairwise distances is modified.

5.1.2 Input Data

The implementation requires two primary input files.

- An aligned FASTA file containing nucleotide sequences. This file provides sequence identifiers, aligned sequences, abundance information, and the root sequence required by ClonalTree.
- A feature table in CSV format. Each row corresponds to one sequence, where the first column contains the sequence identifier and the remaining columns contain numerical features extracted from that sequence.

5.1.3 Single-Feature Representation

The initial implementation ranked the available features and selected a single feature for distance computation.

Each sequence was therefore represented by one scalar value:

$$X_i = f(i).$$

The distance between two sequences was calculated as

$$d(i, j) = |f(i) - f(j)|.$$

The resulting distance matrix was then supplied to the standard ClonalTree Minimum Spanning Tree procedure:

```
primMST(adjMatrix, root, labels, abundance).
```

Trimming and revision procedures were applied when necessary.

5.2 Initial Feature-Based Output

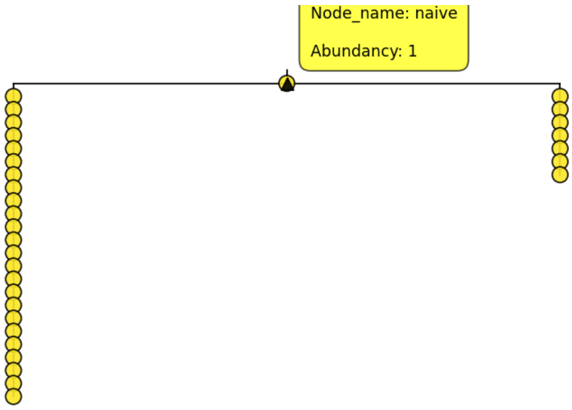


Figure 6: Initial feature-based reconstruction on Dataset 02 using a single-feature distance metric.

5.3 Discussion of the Initial Representation

The initial feature-based implementation successfully generated lineage trees, demonstrating that numerical feature representations can be integrated into the ClonalTree framework.

However, a major limitation became apparent.

Because each sequence was represented by only one numerical value, all sequences were effectively projected onto a one-dimensional space.

In a one-dimensional space, points can be naturally ordered according to their feature values. The Minimum Spanning Tree therefore tends to connect neighboring points sequentially.

This produces a chain-like or stick-like topology.

Figure 6 illustrates this behavior. Most sequences are connected through a dominant linear path, with relatively few branching events.

The resulting topology is less consistent with the expected branching structure of B-cell clonal evolution.

This limitation demonstrates that simply replacing sequence similarity with a numerical feature is not sufficient. The representation must preserve enough information to distinguish different relationships among sequences.

5.4 Comparison with Amino-Acid-Based Reconstruction

The amino-acid-based reconstruction and the initial feature-based reconstruction exhibit substantially different topological behavior.

The amino-acid-based representation preserves positional information across the complete protein sequence. Consequently, differences at multiple positions can contribute to the distance between two sequences.

In contrast, the initial feature-based approach reduces each sequence to a single scalar value.

The dimensionality of the representation therefore provides an important explanation for the difference in topology.

The one-dimensional feature representation strongly constrains the possible relationships between samples and encourages a chain-like MST.

This observation motivated the development of a multidimensional feature-based representation.

5.5 Improved Feature-Based Reconstruction

Several modifications were introduced to overcome the limitations of the initial implementation while preserving the overall ClonalTree framework.

5.5.1 Multi-Feature Distance Representation

The most important modification was replacing the single numerical feature with a multidimensional feature vector.

Each sequence is represented as

$$X_i = (f_1(i), f_2(i), \ldots, f_k(i)).$$

Pairwise distances are calculated using Euclidean distance:

$$d(i, j) = \sqrt{\sum_{m=1}^{k} (X_{i,m} - X_{j,m})^2}.$$

The multidimensional representation allows different aspects of a sequence to contribute simultaneously to the distance.

This provides a richer representation of sequence relationships and reduces the tendency of the MST to collapse into a single linear chain.

5.5.2 Improved Identifier Matching

Sequence identifiers in FASTA files and feature tables are not always formatted identically.

For example,

`seq2852`

and

`seq2852@38`

can correspond to the same biological sequence, with the latter containing additional abundance information.

The revised implementation therefore performs more robust identifier matching.

The matching procedure considers:

- exact identifier matches;
- identifiers after removal of abundance suffixes;
- special handling of the naive/root sequence.

This prevents valid sequences from being discarded because of formatting differences.

5.5.3 Handling Missing Features

The original implementation removed sequences when feature values were unavailable.

The revised implementation instead supports more robust handling of missing entries through feature imputation when appropriate.

This approach helps preserve the expected number of sequences and leaves in the reconstructed lineage tree.

5.5.4 Output Labeling

Abundance labels are appended only after the tree has been fully constructed.

This avoids inconsistencies during distance computation and indexing while preserving the expected ClonalTree output format.

5.6 Improved Feature-Based Output

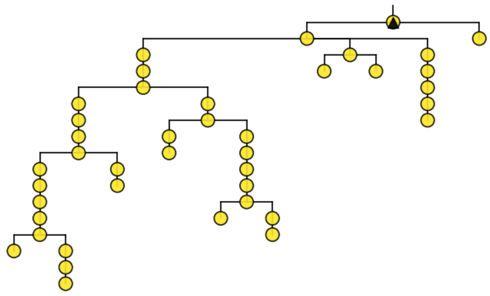


Figure 7: Improved feature-based reconstruction on Dataset 02 using multidimensional feature vectors and Euclidean distance.

5.7 Discussion of the Improved Representation

The improved feature-based reconstruction produces a substantially richer topology than the initial single-feature implementation.

Instead of a single dominant chain, the reconstructed tree contains multiple branching points and several groups of related sequences.

This indicates that the multidimensional feature representation provides additional discriminatory information for the Minimum Spanning Tree construction.

The topology also appears more hierarchical, with branching occurring at multiple levels.

Although elongated branches remain, the strong stick-like behavior observed in the initial implementation is substantially reduced.

This demonstrates that the dimensionality and information content of the distance representation have a major effect on the topology generated by the ClonalTree framework.

5.8 Comparison Between Initial and Improved Feature Representations

The difference between the two feature-based implementations can be summarized as follows.

The initial implementation represents each sequence as

$$X_i = f(i),$$

whereas the improved implementation represents each sequence as

$$X_i = (f_1(i), f_2(i), \dots, f_k(i)).$$

The first representation provides only one degree of freedom.

The second representation allows sequences to differ along multiple dimensions simultaneously.

As a result, the Minimum Spanning Tree has a richer geometric structure from which to determine connections.

The observed transition from a predominantly linear tree to a more branched tree therefore supports the use of multidimensional feature representations for this type of reconstruction.

However, visual improvement in topology should not automatically be interpreted as biological improvement.

As with the amino-acid representation, quantitative validation against known ground-truth trees will ultimately be required to determine whether the feature-based method improves reconstruction accuracy.

6 Biologically Informed Reconstruction

6.1 Motivation

Amino-acid Hamming distance provides a protein-level representation of sequence differences, but it treats every amino-acid substitution equally.

For example, replacing one hydrophobic amino acid with another hydrophobic amino acid contributes the same amount to the Hamming distance as replacing a hydrophobic amino acid with a strongly charged residue.

From a biological perspective, these substitutions may have very different consequences.

This motivates the development of a biologically informed distance measure that incorporates properties of amino acids beyond simple equality or inequality.

6.2 Biological and Physicochemical Properties

Potential properties for incorporation include:

- hydrophobicity;
- polarity;
- charge;
- molecular size;
- side-chain characteristics;
- other chemical or physicochemical descriptors.

Instead of assigning the same distance to every amino-acid substitution, the distance could reflect the similarity between the properties of the two residues.

Conceptually, two residues with similar properties would contribute a smaller distance, while substitutions involving substantially different residues would contribute a larger distance.

6.3 Current Development

The biologically informed representation is currently under active development.

The main objective is to identify a distance formulation that captures biologically meaningful differences while remaining computationally efficient enough for integration into the ClonalTree framework.

The resulting distance matrix will then be supplied to the same MST reconstruction procedure used in the nucleotide-, amino-acid-, and feature-based experiments.

Because this component is still being developed, detailed methodology and experimental results are intentionally not included in the current report.

7 Overall Comparison and Interpretation

The experiments conducted so far demonstrate that the choice of sequence representation has a substantial effect on lineage reconstruction.

The ground-truth validation establishes that the nucleotide-based ClonalTree implementation can recover known lineage relationships with high accuracy.

Across the four simulated datasets, the nucleotide-based method achieved an average F-measure of approximately 0.9508.

This provides a strong baseline against which alternative representations can be evaluated.

The amino-acid-based approach introduces a biologically motivated change by moving from nucleotide-level sequence similarity to protein-level sequence similarity.

This representation removes synonymous nucleotide substitutions from the distance calculation while retaining substitutions that alter the encoded protein sequence.

The feature-based experiments demonstrate a different aspect of the problem.

The initial single-feature implementation produced a predominantly linear topology. This behavior can be explained by the one-dimensional structure of the representation.

The improved multidimensional implementation substantially reduced this limitation and produced more branching structures.

Together, these experiments suggest that the information content and dimensionality of the distance representation are important determinants of the resulting lineage topology.

However, topology alone is not sufficient to establish that one representation is biologically or computationally superior to another.

For this reason, the ground-truth benchmark provides an important common framework for future comparison.

The nucleotide and amino-acid approaches can be evaluated using exactly the same simulated trees, edge definitions, and evaluation metrics.

The feature-based and biologically informed approaches can subsequently be evaluated in the same way.

This will allow the study to distinguish between:

1. representations that merely produce visually different trees;
2. representations that improve quantitative reconstruction accuracy;
3. representations that improve biological interpretability without necessarily improving topology reconstruction.

This distinction is important because a more visually complex or more highly branched tree is not necessarily a more accurate lineage tree.

8 Current Status

At the current stage, the project includes:

- Nucleotide-based ClonalTree reconstruction.
- Generation of four simulated ground-truth lineage trees using Anestra Eco.
- Quantitative validation of nucleotide-based reconstruction against the simulated ground-truth trees.
- Edge-based Precision, Recall, and F-measure evaluation.
- Amino-acid-based ClonalTree reconstruction.
- Visual comparison of ground-truth, nucleotide-based, and amino-acid-based lineage trees.
- Initial single-feature-based ClonalTree reconstruction.
- Improved multidimensional feature-based reconstruction.
- Improved sequence identifier matching.
- Handling of missing feature values.
- Development of biologically informed similarity measures.
- Benchmark evaluation on LLC datasets.

The nucleotide-based ground-truth validation is currently the most complete quantitative component of the study.

The amino-acid-based implementation has been developed and visualized, while its quantitative evaluation against the simulated ground-truth trees remains ongoing.

The feature-based implementation has progressed from a single-feature representation to a multidimensional representation.

The biologically informed representation remains under active development.

9 Current Limitations

Several limitations should be considered when interpreting the current results.

9.1 Limited Number of Simulated Datasets

The current ground-truth validation uses four simulated datasets.

Although these datasets provide useful initial validation, a larger number of simulations would provide stronger evidence about the general performance of the reconstruction pipeline.

9.2 Dependence on Simulation Characteristics

Simulated trees provide a known ground truth, but their characteristics may not fully represent the complexity of real B-cell lineage evolution.

Therefore, strong performance on simulated data does not automatically guarantee equivalent performance on experimental datasets.

9.3 Edge-Based Evaluation

The current evaluation focuses on individual edges.

This provides a straightforward and interpretable measure of topology agreement, but it does not capture every possible aspect of tree similarity.

Additional tree-comparison metrics may therefore be useful in future work.

9.4 Amino-Acid Quantitative Evaluation

The amino-acid reconstruction has not yet been fully evaluated using the ground-truth edge-based metrics.

Therefore, no conclusion should currently be made that amino-acid-based reconstruction is superior or inferior to nucleotide-based reconstruction.

9.5 Feature Scaling

The multidimensional feature-based approach uses numerical features that may have different scales and distributions.

The choice of normalization or standardization can therefore influence the resulting Euclidean distances.

Further investigation is required to determine the most appropriate feature preprocessing strategy.

10 Future Work

Future work will focus on several directions.

10.1 Quantitative Amino-Acid Evaluation

The first priority is to apply the ground-truth evaluation procedure to the amino-acid-based reconstructions.

Precision, Recall, and F-measure will be calculated for all four simulated datasets.

The results will then be directly compared with the nucleotide-based baseline.

10.2 Direct DNA versus Amino-Acid Comparison

The quantitative comparison will determine whether amino-acid-based distance preserves, improves, or reduces reconstruction accuracy relative to nucleotide Hamming distance.

The comparison will be performed using the same ground-truth trees and evaluation procedure.

10.3 Biologically Informed Distance Measures

Further work will investigate distance functions that incorporate physicochemical properties of amino acids.

Potential approaches include representing each amino acid using a vector of properties and calculating distances between these vectors.

This would provide a continuous measure of biochemical similarity rather than the binary distinction used by Hamming distance.

10.4 Feature Selection and Normalization

The multidimensional feature-based approach will be further investigated by testing different feature subsets and normalization strategies.

This will help determine whether certain features contribute more strongly to lineage reconstruction quality.

10.5 Additional Simulated Datasets

Additional ground-truth datasets will be generated to test whether the observed reconstruction performance generalizes across different tree sizes, branching patterns, mutation rates, and sequence characteristics.

10.6 Biological Validation

Ultimately, the reconstructed trees should be evaluated not only using topological similarity but also according to biological plausibility.

This may include examining known clonal relationships, mutation patterns, and other biological characteristics of the underlying B-cell populations.

10.7 Computational Efficiency

An additional objective is to preserve the computational efficiency of the original ClonalTree framework.

Alternative distance representations should therefore be evaluated not only for reconstruction quality but also for computational cost and scalability.

10.8 Publication Preparation

After completing the quantitative comparisons and biologically informed methods, the results will be consolidated into a complete study suitable for future publication.

11 Conclusion

This report presents ongoing work investigating alternative representations for B-cell lineage reconstruction using the ClonalTree framework.

The study began by establishing a nucleotide-based baseline and validating the reconstruction pipeline against four simulated lineage trees generated using Anestra Eco.

The ground-truth evaluation demonstrated strong reconstruction performance. The nucleotide-based implementation achieved F-measures of 0.9783, 0.9390, 0.9169, and 0.9688 across the four datasets, with an average F-measure of approximately 0.9508.

These results indicate that the baseline implementation successfully recovers the majority of known lineage relationships.

An amino-acid-based implementation was subsequently developed to investigate whether protein-level sequence representation provides an alternative view of lineage relationships.

The amino-acid approach uses IMGT-derived protein sequences and amino-acid Hamming distance while preserving the underlying ClonalTree MST procedure.

Feature-based reconstruction was also investigated.

The initial single-feature representation produced predominantly linear topologies, demonstrating the limitations of reducing each sequence to a single numerical descriptor.

An improved multidimensional representation substantially increased the complexity and branching structure of the resulting trees, demonstrating the importance of representation richness in distance-based reconstruction.

Finally, biologically informed reconstruction is being developed to incorporate chemical and physicochemical properties of amino acids.

The next major step is to quantitatively evaluate the amino-acid and feature-based approaches against the same ground-truth datasets used for the nucleotide baseline.

This will allow the different representations to be compared objectively and will determine whether protein-level or biologically informed distance measures can improve lineage reconstruction while maintaining the efficiency of the ClonalTree framework.

This report summarizes ongoing research. Certain implementation details, algorithms, and experimental analyses have been intentionally omitted because the work is still in progress and is being prepared for future publication.